

Graduate Algebra

September 27, 2026

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1. Week 1 [2026-09-09]

We'll begin by recalling some basic sorts of algebra that you more-or-less encountered before.

1.1. Notations and recollections

We reserve the following letters:

- $\mathbb{N}$ for the set of *natural numbers* $0, 1, 2, \dots$
- $\mathbb{Z}$ for the set of *integers*, i.e. for all $\pm n$ for $n \in \mathbb{N}$
- $\mathbb{Q}$ for the set of *rational numbers* m/n for $m, n \in \mathbb{Z}$ with $n \neq 0$
- $\mathbb{R}$ for the set of *real numbers*, and
- $\mathbb{C}$ for the set of *complex numbers* $a + bi$ for $a, b \in \mathbb{R}$.

In this first lecture, I want to recall some of the main objects of study in algebra, including: groups, rings and fields. Ultimately, the goal today is to prove an analogue of Cayley's Theorem - see Theorem 1.6.1 and Theorem 1.7.1 about embedding arbitrary groups in some standard groups.

1.2. Groups

Recall that a group is a set G together with a binary operation $\cdot : G \times G \rightarrow G$ satisfying the following:

- associativity: $\forall x, y, z \in G, (xy)z = x(yz)$
- identity: $\exists e \in G, xe = ex = x$.
- inverses: $\forall x \in G, \exists y \in G, xy = yx = 1$.

Remark 1.2.1:

- We usually write 1 or sometimes 1_G rather than e for the inverse element of G .
- we usually write x^{-1} for the inverse of $x \in G$
- there are *uniqueness* results that I'm eliding here; the identity 1 of G is unique, and the inverse x^{-1} of an element is unique. These statements are *consequences* of the above axioms (they don't require additional assumption.)
- A group is abelian if $\forall a, b \in G, ab = ba$
- Sometimes we write groups additively; in that case, 0 is the identity element and the inverse of $a \in G$ is $-a \in G$. We always insist that additive groups are abelian.

Definition 1.2.2: For groups G and H , a function $\varphi : G \rightarrow H$ is a **group homomorphism** provided that $\forall x, y \in G, \varphi(xy) = \varphi(x)\varphi(y)$.

Definition 1.2.3: Let $\varphi : G \rightarrow H$ be a group homomorphism. The **kernel** of φ is

$$\ker \varphi = \{g \in G \mid \varphi(g) = 1.\}$$

Remark 1.2.4: If $\varphi : G \rightarrow H$ is a group homomorphism, $\ker \varphi$ is a subgroup of G - i.e. $\ker \varphi$ is non-empty, and is closed under multiplication and under taking inverses.

Proposition 1.2.5: Let $\varphi : G \rightarrow H$ be a group homomorphism. Then φ is an injective (or one-to-one) function if and only if $\ker \varphi = \{1_G\}$.

1.3. Rings

Definition 1.3.1: A **ring** is an additive abelian group R together with a binary operation of multiplication

$$\cdot : R \times R \rightarrow R$$

which satisfies the following:

- multiplication is associative: $\forall a, b, c \in R, (ab)c = a(bc)$.
- there is a multiplicative identity: $\exists 1 \in R, \forall a \in R, 1a = a1 = a$.
- distribution laws: $\forall a, b, c \in R, a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

The ring R is **commutative** provided that $\forall a, b \in R, ab = ba$.

Example 1.3.2:

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are commutative rings.
- For a natural number $n > 1$, the ring $\text{Mat}_n(\mathbb{Z})$ of $n \times n$ matrices with coefficients in $\mathbb{Z}$ is a non-commutative ring.

Definition 1.3.3: For a commutative ring R , an element $a \in R$ is a **unit** provided that $\exists v \in R, uv = vu = 1$.

The set $R^\times$ of units in R is a group under the multiplication of R .

1.4. Fields

Definition 1.4.1: A **field** is a commutative ring F such that $\forall a \in F, a \neq 0 \Rightarrow a$ is a unit.

Example 1.4.2: $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields, but $\mathbb{Z}$ is not a field.

Remark 1.4.3: A possibly non-commutative ring satisfying the requirement $\forall a \in F, a \neq 0 \Rightarrow a$ is a unit is known as a **division ring**. The most well-known example of a non-commutative division ring is Hamilton's quaternions $\mathbb{H}$: $\mathbb{H}$ is a 4 dimensional $\mathbb{R}$ -vector space with basis $1, i, j, k$ where the basis vectors multiply according to the rules $i^2 = j^2 = k^2 = -1, ij = -ji = k, jk = -kj = i$ and $ki = -ik = j$.

If $0 \neq v = a + bi + cj + dk \in \mathbb{H}$, then $v^{-1} = (a - bi - cj - dk)/(a^2 + b^2 + c^2 + d^2)$.

1.5. Linear Algebra

Definition 1.5.1: If F is a field, a vector space over F – or an F -vector space – is an additive abelian group V together with an operation of scalar multiplication

$$F \times V \rightarrow V$$

written $(t, v) \mapsto tv$, subject to the following:

- identity: $\forall v \in V, 1v = v$.
- associativity: $\forall a, b \in F$ and $v \in V, a(bv) = (ab)v$.
- distributive laws: $\forall a, b \in F$ and $v, w \in V, (a + b)v = av + bv$ and $a(v + w) = av + aw$.

Remark 1.5.2: Probably in a linear algebra class you saw results stated for vector spaces over $\mathbb{R}$ or $\mathbb{C}$; however, many general results in linear algebra remain valid for vector space over F .

Example 1.5.3: Let I be any set, and let V be the set of all functions $f : I \rightarrow F$ which have finite support. Recall that the support of f is $\{x \in I \mid f(x) \neq 0\}$.

Then V is a vector space. (The addition and scalar multiplication operations are define “pointwise” – see homework.)

Remark 1.5.4: Recall that a **basis** of a vector space is subset B of V which is linearly indepent and spans V . The vector space of finitely supported functions $I \rightarrow F$ has a basis $B = \{\delta_i \mid i \in I\}$, where

$$\delta_i : I \rightarrow F$$

is the function defined by $\delta_i(j) = 0$ if $i \neq j$ and $\delta_i(i) = 1$.

Definition 1.5.5: If V and W are F -vector spaces, an F -linear map $\varphi : V \rightarrow W$ is a homomorphism of additive groups which satisfies the condition

$$\forall t \in F, \forall v \in V, \varphi(tv) = t\varphi(v).$$

Definition 1.5.6: If V is an F -vector space, the general linear group $\text{GL}(V)$ is the set

$$\{\varphi : V \rightarrow V \mid \varphi \text{ is } F\text{-linear and invertible.}\}$$

$\text{GL}(V)$ is a group whose operation is given by composition of linear transformations.

Remark 1.5.7: If V is finite dimensional, so that V is isomorphic to F^n as F -vector spaces, linear algebra shows that $\text{GL}(V)$ is isomorphic to the group GL_n of $n \times n$ matrices with non-zero determinant, where $n = \dim_F V$ and where the operation in GL_n is given by matrix multiplication.

Example 1.5.8: If I is a finite set with $n \in \mathbb{N}$ elements, then the vector space of finitely supported F -valued functions on I is isomorphic to the space F^n of column vectors.

Remark 1.5.9: In fact, if D is a division ring one can consider vector spaces over D . Rather than make an ad hoc definition for this, we will later define the notion of a module over a ring. A vector space over a field F is just an F -module, and then a vector space over a division algebra D is just a D -module.

1.6. Cayley's Theorem

Let Ω be any set. The set $S(\Omega)$ of all bijective functions $\psi : \Omega \rightarrow \Omega$ is a group whose operation is composition of functions.

Theorem 1.6.1 (Cayley's Theorem): Let G be any group. Then G is isomorphic to a subgroup of $S(\Omega)$ for some Ω .

Proof: Let $\Omega = G$. For $g \in G$, define a mapping $\lambda_g : G \rightarrow G$ by the rule

$$\lambda_g(h) = gh.$$

We are going to argue that the mapping $g \mapsto \lambda_g$ defines an injective group homomorphism

$$G \rightarrow S(\Omega) = S(G).$$

First of all, we note that $\lambda_1 = \text{id}$. Indeed, to check this identity of functions, let $h \in \Omega = G$. Then

$$\lambda_1(h) = 1h = h = \text{id}(h);$$

this confirms $\lambda_1 = \text{id}$.

Next, we note that for $g_1, g_2 \in G$, we have $(*) \quad \lambda_{g_1} \circ \lambda_{g_2} = \lambda_{g_1 g_2}$. Again, to confirm this identity of functions, we let $h \in \Omega = G$. Then

$$(\lambda_{g_1} \circ \lambda_{g_2})h = \lambda_{g_1}(\lambda_{g_2}(h)) = \lambda_{g_1}(g_2 h) = g_1(g_2 h) = (g_1 g_2)h = \lambda_{g_1 g_2}(h)$$

as required.

Now, using $(*)$ we see for $g \in G$ that $\lambda_g \circ \lambda_{g^{-1}} = \lambda_1 = \text{id} = \lambda_{g^{-1}} \circ \lambda_g$, which proves that λ_g is bijective; thus indeed $\lambda_g \in S(\Omega) = S(G)$.

Moreover, $(*)$ shows that the mapping $\lambda : G \rightarrow S(G)$ given by $g \mapsto \lambda_g$ is a group homomorphism.

It remains to see that λ is injective. If $g \in \ker \lambda$, then $\lambda_g = \text{id}$. Thus $1 = \text{id}(1) = \lambda_g(1) = g1 = g$. Thus $g = 1$ so that $\ker \lambda = \{1\}$ which confirms that λ is injective by Proposition 1.2.5. This completes the proof. ■

1.7. A linear analogue of Cayley's Theorem.

Let F be a field.

Theorem 1.7.1: Let G be any group. Then G is isomorphic to a subgroup of $\text{GL}(V)$ for some F -vector space V .

Proof: The proof is quite similar to the proof of Cayley's Theorem.

Let V be the vector space of all finitely supported functions $f : G \rightarrow F$. Recall that V has a basis $B = \{\delta_g \mid g \in G.\}$

We are going to define an injective group homomorphism $G \rightarrow \text{GL}(V)$.

For $g \in G$, we may define an F -linear mapping $\lambda_g : V \rightarrow V$ by specifying the value of λ_g at each vector in B . We set $\lambda_g(\delta_h) = \delta_{gh}$ for each $h \in G$.

Recall that a typical element v of V has the form

$$v = \sum_{i=1}^n t_i \delta_{h_i}$$

for scalars $t_i \in F$ and elements $h_i \in G$; since λ_g is F -linear, we have

$$\lambda_g(v) = \sum_{i=1}^n t_i \delta_{gh_i}.$$

We now show that $\lambda_1 = \text{id}$. To prove this, since the functions $V \rightarrow V$ are linear, it is enough to argue that the functions agree at each element of the basis B of V . Well, for $h \in G$,

$$\lambda_1(\delta_h) = \delta_{1h} = \delta_h = \text{id}(\delta_h)$$

as required.

We next show for $g_1, g_2 \in G$ that $(*) \quad \lambda_{g_1} \circ \lambda_{g_2} = \lambda_{g_1 g_2}$. Again, it suffices to argue that these functions agree at each element δ_h of B . For $h \in G$ we have:

$$(\lambda_{g_1} \circ \lambda_{g_2})(\delta_h) = \lambda_{g_1}(\lambda_{g_2} \delta_h) = \lambda_{g_1}(\delta_{g_2 h}) = \delta_{g_1(g_2 h)} = \delta_{(g_1 g_2)h} = \lambda_{g_1 g_2} \delta_h$$

as required.

Now, for $g \in G$ we see that by $(*)$ that

$$\text{id} = \lambda_1 = \lambda_g \circ \lambda_{g^{-1}}$$

which proves that λ_g is invertible and hence in $\text{GL}(V)$.

Moreover, (*) shows that the assignment $\lambda : G \rightarrow \text{GL}(V)$ given by the rule $g \mapsto \lambda_g$ is a group homomorphism.

It remains to argue that λ is injective. Suppose that $x \in \ker \lambda$, so that $\text{id} = \lambda_x$.

Then $\delta_1 = \text{id}(\delta_1) = \lambda_x(\delta_1) = \delta_{x1} = \delta_x$. This implies that $1 = x$ so that indeed the kernel of λ is trivial and thus λ is injective by Proposition 1.2.5.

■

2. Week 2 [2026-09-14]

2.1. Quotient of a set by an equivalence relation

Let S be a set and let $\sim$ be an equivalence relation on S – recall this means $\sim$ is **reflexive**, **symmetric**, and **transitive**: $\forall s, s \sim s$; $s_1 \sim s_2 \Rightarrow s_2 \sim s_1$; and $s_1 \sim s_2$ and $s_2 \sim s_3 \Rightarrow s_1 \sim s_3$.

Definition 2.1.1: For $s \in S$, the **equivalence class** $[s]$ is the subset

$$[s] = \{x \in S \mid x \sim s\}.$$

Proposition 2.1.2: For arbitrary $s, s' \in S$:

- a. $s \sim s' \Leftrightarrow [s] = [s']$
- b. $[s] \neq [s'] \Leftrightarrow [s] \cap [s'] = \emptyset$

Proof: Review! ■

Proposition 2.1.3: S is the disjoint union of its equivalence classes.

Proof: Each $x \in S$ lies in $[x]$. If $z \in [x] \cap [y]$ then $x \sim z$ and $y \sim z$, so $x \sim y$ by transitivity and $[x] = [y]$. ■

Definition 2.1.4: The **quotient** of S by $\sim$ is the pair $(\bar{S}, \pi)$ where $\bar{S} = \{[s] \mid s \in S\}$ and $\pi : S \rightarrow \bar{S}$ is given by $\pi(s) = [s]$.

By construction, π is surjective, and Proposition 2.1.2 (a) gives $a \sim b \Rightarrow \pi(a) = \pi(b)$ (with equality forced **only** in that case).

Remark 2.1.5: For $x \in S$, $\pi^{-1}(\pi(x)) = [x]$: this is immediate from Proposition 2.1.2 (a).

Proposition 2.1.6: (Universal property of the quotient) Let T be a set and $f : S \rightarrow T$ a function with $\forall a, b \in S, a \sim b \Rightarrow f(a) = f(b)$. Then there is a **unique** function $\bar{f} : \bar{S} \rightarrow T$ with $f = \bar{f} \circ \pi$.

Proof: Proposition 2.1.2 (a) shows the rule $[s] \mapsto f(s)$ is well-defined, giving $\bar{f}$ with $\bar{f} \circ \pi = f$. Since π is surjective, any function satisfying this equation already agrees with $\bar{f}$ everywhere, so $\bar{f}$ is unique. ■

Remark 2.1.7: Proposition 2.1.6 is really what characterizes $(\bar{S}, \pi)$: applying it in both directions to two quotients of the same $\sim$ produces mutually inverse maps between them, so any two quotients are canonically isomorphic. (Good exercise – write this out.) It's harmless, then, to work with **the** quotient $S/\sim$ via whichever concrete model is convenient – here, equivalence classes.

Readers who've seen Lean's Quot/Quotient type will recognize this shape: `Quot.mk` is π , `Quot.lift` builds $\bar{f}$ from f plus a proof that f respects $\sim$, and `Quot.sound` is Proposition 2.1.2 (a).

2.2. Sub-groups

Let G be a group (when giving definitions, we'll write G multiplicatively).

Definition 2.2.1: A **subgroup** of G is a non-empty subset $H \subseteq G$ such that H is closed under the operations of multiplication in G and inversion in G . In other words,

$$\forall a, b \in G, ab \in H \text{ and } a^{-1} \in H$$

.

Example 2.2.2: Consider the group $G = \mathbb{Z} \times \mathbb{Z}$ where the operation is componentwise addition. Check the following!

- $H_1 = \{(a, b) \in G \mid 2a + 3b = 0\}$ is a subgroup.
- $H_2 = \{n(2, 2) + m(1, 2) \mid n, m \in \mathbb{Z}\}$ is a subgroup.

The collection of subgroups of G has a natural partial order given by *containment*.

Proposition 2.2.3: (Constructing subgroups)

- If H_i for $i \in I$ is a family of subgroups of G , indexed by some set I , then the intersection $\bigcap_{i \in I} H_i$ is again a subgroup of G .
- Let $S \subseteq G$ be a subset. There is a unique smallest subgroup $H(S) = \langle S \rangle$ containing S . In other words, for any subgroup H' of G with $S \subseteq H'$, we have $\langle S \rangle \subseteq H'$.

Remark 2.2.4:

- If $S, T \subseteq G$ are subsets, we often write $\langle S, T \rangle$ for $\langle S \cup T \rangle$. If $S = \{s_1, s_2, \dots, s_n\}$ we often write $\langle S \rangle = \langle s_1, s_2, \dots, s_n \rangle$.
- The subgroup in Example 2.2.2(b) is precisely $\langle (2, 2), (1, 2) \rangle$.
- For any group G and $a \in G$, $\langle a \rangle := \langle \{a\} \rangle$ is the **cyclic subgroup** generated by a . If G is multiplicative, then $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$ while if G is additive then $\langle a \rangle = \{na \mid n \in \mathbb{Z}\}$.

2.3. Group actions

Definition 2.3.1: Let G be a group and let Ω be a set. An **action** of G on Ω is a mapping

$$G \times \Omega \rightarrow \Omega \text{ written } (g, x) \mapsto gx$$

such that for each $x \in \Omega$ we have

- $1x = x$
- $\forall g, h \in G, (gh)x = g(hx)$.

For brevity, sometimes we say that Ω is a G -space.

Proposition 2.3.2: An action of a group G on a set Ω determines a homomorphism $f : G \rightarrow S(\Omega)$ such that $f(g)(x) = gx$ for $g \in G$ and $x \in \Omega$.

Conversely, given a homomorphism $f : G \rightarrow S(\Omega)$, there is an action of G on Ω given by $gx = f(g)(x)$ for each $g \in G$ and $x \in \Omega$.

Definition 2.3.3: Suppose that Ω is a G -space. The G -conjugacy relation on Ω is defined as follows: for $x, y \in \Omega$, $x \sim_G y$ provided that $\exists g \in G, gx = y$.

Proposition 2.3.4: The G -conjugacy relation on Ω is an equivalence relation.

Definition 2.3.5: Let Ω be a G -space, and let $\varphi : \Omega \rightarrow \Omega / \sim$ be the quotient mapping for the G -conjugacy relation; see Definition 2.1.4. For $x \in \Omega$, the **orbit** $\mathcal{O}_x = Gx$ of G through x is the subset of Ω defined by

$$\mathcal{O}_x = \varphi^{-1}(\varphi(x)).$$

Thus the G -orbits are the equivalence classes for the relation $\sim_G$; see Remark 2.1.5.

Equivalently, we have $\mathcal{O}_x = \{gx \mid g \in G.\}$

Proposition 2.3.6: Ω is the disjoint union of the G -orbits in Ω .

Proof: This follows from Proposition 2.1.3. ■

Remark 2.3.7: Each orbit $\mathcal{O}_x$ is itself a G -set.

2.4. Quotients of groups

Let G be a group and let H be a subgroup of G . There is an action of H on the set G by right multiplication: for $h \in H$ and $g \in G$ we can define $h \cdot g = gh^{-1}$.

We are going to consider the quotient of G by the equivalence relation of H -conjugacy; this equivalence relation is defined by

$$g \sim g' \Leftrightarrow \exists h \in H, g = g'h.$$

Definition 2.4.1: The **left quotient of G by H** is the quotient $(\pi, G/H)$ of G by the equivalence relation of H -conjugacy defined using the action of H on G by right multiplication as described above.

Remark 2.4.2:

- Of course, one can use an explicit model for the quotient by taking G/H to be the set of equivalence classes in G for the H -conjugacy relation.
- The equivalence classes for the relation of H -conjugacy defined by the action of right multiplication are precisely the **left cosets** of H in G . The class of $x \in G$ has the form

$$xH = \{xh \mid h \in H\}$$

.

For $x \in G$,

$$\pi^{-1}(\pi(x)) = xH.$$

- We can also consider the action of H on G by left multiplication. This action determines an equivalence relation of H -conjugacy, and the quotient of G by this equivalence relation is called the **right quotient of G by H** and is written $(\pi, H \backslash G)$. In this case, the equivalence classes are the **right cosets** where the class of $x \in G$ has the form $Hx = \{hx \mid h \in H\}$.

For $x \in G$, we have $\pi^{-1}(\pi(x)) = Hx$.

Proposition 2.4.3: There is an action

$$\alpha : G \times G/H \rightarrow G/H$$

of the group G on the set G/H such that

$$\forall g, x \in G, \text{ we have } \alpha(g, \pi(x)) = \pi(gx)$$

where $\pi : G \rightarrow G/H$ is the quotient map.

Proof: To define the action map α , first fix $g \in G$. We are going to define the mapping

$$\alpha(g, -) : G/H \rightarrow G/H.$$

Consider the mapping $\pi_g : G \rightarrow G/H$ given by the rule $\pi_g(x) = \pi(gx)$. This mapping has the property that $x \sim_H x' \Rightarrow \pi_g(x) = \pi_g(x')$. Indeed,

$$x \sim_H x' \Rightarrow \exists h, x = x'h \Rightarrow \pi_g(x) = \pi(gx) = \pi(gx'h) = \pi(gx') = \pi_g(x')$$

by the defining property of π ; see Definition 2.1.4. Again using Definition 2.1.4 we find the desired mapping $\alpha(g, -) : G/H \rightarrow G/H$ with the property that

$$(\clubsuit) \quad \alpha(g, -) \circ \pi = \pi_g.$$

We now assemble the mappings $\alpha(g, -)$ to get a mapping $\alpha : G \times G/H \rightarrow G/H$ which satisfies $\alpha(g, \pi(x)) = \pi(gx)$ for each $g, x \in G$, and it remains to check that α determines an action as in Definition 2.3.1.

Of course, using $(\clubsuit)$, we have $\alpha(1, -) \circ \pi = \pi_1 = \pi = \text{id} \circ \pi$; since π is surjective, it follows that $\alpha(1, -) = \text{id}$. Thus $\alpha(1, z) = z$ for each $z \in G/H$, which shows that α satisfies the first requirement of Definition 2.3.1.

Now suppose that $g_1, g_2 \in G$. To complete the proof, we must verify the remaining requirement of Definition 2.3.1; thus we must show that

$$(\heartsuit) \quad \alpha(g_1, \alpha(g_2, -)) = \alpha(g_1 g_2, -)$$

On the one hand, using $(\clubsuit)$ we find that

$$\alpha(g_1 g_2, -) \circ \pi = \pi_{g_1 g_2};$$

on the other hand, for $z \in G$ we have

$$\begin{aligned} (\alpha(g_1, \alpha(g_2, -)) \circ \pi)(z) &= \alpha(g_1, \alpha(g_2, \pi(z))) \\ &= \alpha(g_1, \pi_{g_2}(z)) && \text{by } (\clubsuit) \\ &= \alpha(g_1, \pi(g_2 z)) \\ &= \pi(g_1(g_2 z)) && \text{by } (\clubsuit) \end{aligned}$$

Since π is surjective, $(\heartsuit)$ follows at once. This completes the proof. ■

2.5. Quotients of groups and orbits.

Definition 2.5.1: Suppose that G acts on Ω_1 and on Ω_2 . A morphism of G -sets $\varphi : \Omega_1 \rightarrow \Omega_2$ is a function φ with the property that $\forall g \in G$ and $\forall x \in \Omega_1$, we have $\varphi(gx) = g\varphi(x)$.

The morphism of G -sets φ is an isomorphism (of G -sets) if there is a morphism of G -sets $\psi : \Omega_2 \rightarrow \Omega_1$ such that $\varphi \circ \psi = \text{id}$ and $\psi \circ \varphi = \text{id}$.

Suppose that G acts on Ω and let $x \in \Omega$.

Definition 2.5.2: The **stabilizer of x in G** is the subgroup $\text{Stab}_G(x) = \{g \in G \mid gx = x\}$.

Proposition 2.5.3: Write $H = \text{Stab}_G(x)$ and recall that $\pi : G \rightarrow G/H$ is the quotient mapping. There is a unique isomorphism of G -sets $\gamma : G/H \rightarrow \mathcal{O}_x$ with the property that

$$\gamma(\pi(1)) = x.$$

Proof: The rule $g \mapsto gx$ determines a surjective mapping $\alpha_x : G \rightarrow \mathcal{O}_x$. Recall that the action of H on G by right multiplication determines an equivalence relation $\sim$ on G used to construct the quotient G/H .

For $g_1, g_2 \in G$ we find that

$$g_1 \sim g_2 \Rightarrow \exists h \in H, g_1 = g_2 h \Rightarrow \alpha(g_1) = \alpha(g_2 h) = g_2 h x = g_2 x = \alpha(g_2)$$

since $h \in H = \text{Stab}_G(x) \Rightarrow hx = x$.

Thus Definition 2.1.4 shows that there is a mapping $\gamma : G/H \rightarrow \mathcal{O}_x$ such that $\gamma \circ \pi = \alpha_x$. To see that γ is a morphism of G -sets, it suffices to show that $(\clubsuit) \quad \forall g, g' \text{ we have}$

$$\gamma(g \cdot \pi(g')) = g \cdot \gamma(\pi(g')).$$

Now by the definition of the G -action on G/H we have $g \cdot \pi(g') = \pi(gg')$; see Proposition 2.4.3. Thus $\gamma(g \cdot \pi(g')) = \gamma(\pi(gg')) = \alpha_x(gg') = gg' \cdot x$. On the other hand, $g \cdot \gamma(\pi(g')) = g \cdot \alpha_{x(g')} = g \cdot g' \cdot x$ which confirms $(\clubsuit)$. This shows that γ is indeed a morphism of G -sets.

Since α_x is surjective and $\gamma \circ \pi = \alpha_x$, also γ is surjective. It only remains to see that γ is injective. Suppose that $z, z' \in G/H$ such that $\gamma(z) = \gamma(z')$. Since $\pi : G \rightarrow G/H$ is surjective, we may choose $g, g' \in G$ with $z = \pi(g)$ and $z' = \pi(g')$. Now

$$\gamma(z) = \gamma(z') \Rightarrow \gamma(\pi(g)) = \gamma(\pi(g')) \Rightarrow \alpha_{x(g)} = \alpha_{x(g')} \Rightarrow gx = g'x.$$

We now conclude that $g^{-1}gx = x$ so that $g^{-1}g \in \text{Stab}_G(x) = H$. Since the quotient mapping π is constant on H -orbits, $z = \pi(g) = \pi(gg^{-1}g') = \pi(g') = z'$. This shows that γ is injective and completes the proof. ■

Definition 2.5.4: The action of G on Ω is **transitive** if there is a single G -orbit on Ω . Equivalently, the action is transitive if the quotient $\Omega / \sim$ is a singleton set.

Example 2.5.5: Let I be a set and let $G = S(I)$ be the group of permutations of I . Fix $x \in I$ and let $H = \text{Stab}_G(x)$. Notice that G acts on I . Moreover, the G -orbit of x is precisely I - in other words, the action of G on I is transitive.

Notice that $H = S(I - \{x\})$.

Now Proposition 2.5.3 gives an isomorphism of G -sets $G/H \rightarrow I$; i.e. $S(I)/S(I - \{x\}) \rightarrow I$.

2.6. Products of subgroups and symmetric groups

Definition 2.6.1: If $H, K \subseteq G$ are subgroups, H **normalizes** K if for each $g \in H$ we have $gKg^{-1} \subseteq K$, where $gKg^{-1} = \{gxg^{-1} \mid x \in K\}$.

Definition 2.6.2: Let H, K be subsets of G . The **product** of H and K is

$$HK := \{xy \mid x \in H, y \in K\}.$$

For homework, you will prove

Proposition 2.6.3: Suppose that H, K are subgroups of G and that H normalizes K . Then $\langle H, K \rangle = HK$. In particular, HK is a subgroup of G .

Remark 2.6.4: It's often useful to view the symmetric group S_n as $S(\mathbb{Z}/n\mathbb{Z})$, the permutations of the integers mod n . Once you do, you get to **compute** with indices using ordinary arithmetic in $\mathbb{Z}/n\mathbb{Z}$ – addition, negation, etc. – rather than tracking permutations by hand. This can be a useful point-of-view.

Example 2.6.5: View $S = S_n$ as $S(\mathbb{Z}/n\mathbb{Z})$. Define $\sigma, \tau \in S$ by

$$\sigma(i) = i + 1 \text{ and } \tau(i) = -i,$$

with the arithmetic taking place in $\mathbb{Z}/n\mathbb{Z}$. As permutations, σ is the n -cycle $(1, 2, \dots, n)$ and τ is the product of transpositions $\prod_{i=1}^{\lfloor \frac{n}{2} \rfloor} (i, n - i)$; in particular σ has order n and τ has order 2.

We check the normalizer condition $\tau\langle\sigma\rangle\tau^{-1} \subseteq \langle\sigma\rangle$ directly: for $i \in \mathbb{Z}/n\mathbb{Z}$,

$$(\tau\sigma\tau)(i) = \tau(\sigma(\tau(i))) = \tau(\sigma(-i)) = \tau(-i + 1) = i - 1 = \sigma^{-1}(i),$$

so $\tau\sigma\tau = \sigma^{-1} \in \langle\sigma\rangle$. Thus $\langle\tau\rangle$ normalizes $\langle\sigma\rangle$, and Proposition 2.6.3 shows

$$\langle\sigma, \tau\rangle = \langle\sigma\rangle\langle\tau\rangle$$

is a subgroup of S . We call $D = \langle\sigma, \tau\rangle$ the **dihedral group** (of order to be determined – see the problem set).

2.7. Lagrange's Theorem

Let H be a subgroup of the group G and write G/H for the (left) quotient, as above. Recall that the H -cosets xH are the H -orbits for this action.

Theorem 2.7.1: There is a bijection $\varphi : (G/H) \times H \rightarrow G$ for which $\{\varphi(z, h) \mid h \in H\}$ is an H -orbit (i.e. a left H -coset) for each $z \in G/H$.

Proof: Indeed, using the axiom of choice we select for each $z \in G/H$ an element $g_z \in \pi^{-1}(z)$ where $\pi : G \rightarrow G/H$ is the quotient map.

Now define $\varphi : (G/H) \times H \rightarrow G$ by the rule $\varphi(z, h) = g_z h$.

To see that φ is onto, let $g \in G$. One then knows that $g \sim g_z$ for some $z \in G/H$. Since $\pi^{-1}(z) = g_z H$ it follows that $g = g_z h$ for some $h \in H$, so $g = \varphi(z, h)$.

To see that φ is injective, suppose that $\varphi(z, h) = \varphi(z', h')$. Then $g_z h = g_{z'} h'$ so that

$$(g_{z'})^{-1} g_z \in H \Rightarrow g_z \sim g_{z'} \Rightarrow z = z'.$$

Now $g_z h = g_z h' \Rightarrow h = h'$ which completes the proof that φ is injective. The remaining assertion follows from the definition of φ . ■

Corollary 2.7.2: Suppose that G is a finite group and that H is a subgroup of G . Then

$$|G| = |G/H| \cdot |H|.$$

Proof: Indeed, for finite sets X and Y , we have $|X \times Y| = |X| |Y|$. ■

3. Week 3 [2026-09-21]

3.1. Normal subgroups

Subgroups of the form $\ker \varphi$ have a property that ordinary subgroups might lack; in this section we describe this property.

Proposition 3.1.1: Let G be a group.

- a. For $g \in G$, the assignment $x \mapsto gxg^{-1}$ determines a group isomorphism

$$\text{Inn}_g : G \rightarrow G.$$

- b. The assignment $g \mapsto \text{Inn}_g$ determines a group homomorphism $\text{Inn} : G \rightarrow \text{Aut}(G)$ where $\text{Aut}(G)$ is the group of *automorphisms* of G .

Proof sketch:

- First check that Inn_g is a group homomorphism.
- Then check that $(\blacklozenge) \quad \text{Inn}_g \circ \text{Inn}_h = \text{Inn}_{gh}$ for all $g, h \in G$.
- Next, check that $\text{Inn}_1 = \text{id}$. Using $(\blacklozenge)$, this shows that $(\text{Inn}_g)^{-1} = \text{Inn}_{g^{-1}}$ so indeed Inn_g is an *automorphism* of G .
- Finally, $(\blacklozenge)$ shows that Inn is a group homomorphism.

■

Definition 3.1.2: A subset $N \subseteq G$ is a **normal subgroup** of G if N is a subgroup of G and if for any $g \in G$ and for any $x \in N$, we have $gxg^{-1} \in N$.

Using earlier notation, a subgroup N is normal if $\forall g \in G, \text{Inn}_g N \subseteq N$.

Remark 3.1.3: If N is a normal subgroup then for every $g \in G$ we have $\text{Inn}_g N = N$.

Indeed, normality gives $\text{Inn}_g N \subseteq N$ for every g . Applying this to g^{-1} gives $\text{Inn}_{g^{-1}} N \subseteq N$, and applying Inn_g to both sides gives

$$N = \text{Inn}_g(\text{Inn}_{g^{-1}} N) \subseteq \text{Inn}_g N.$$

Thus $\text{Inn}_g N \subseteq N \subseteq \text{Inn}_g N$, which confirms the assertion.

Definition 3.1.4: For a subgroup K of G , the **normalizer** of K in G is

$$N_G(K) = \{g \in G \mid \text{Inn}_g K = K\}.$$

Note that $N_G(K)$ is a subgroup of G : it contains 1; if $g, g' \in N_G(K)$ then $\text{Inn}_{gg'} K = \text{Inn}_g(\text{Inn}_{g'} K) = \text{Inn}_g K = K$; and if $g \in N_G(K)$ then $\text{Inn}_{g^{-1}} K = \text{Inn}_{g^{-1}}(\text{Inn}_g K) = K$. Moreover $K \subseteq N_G(K)$, and K is a normal subgroup of $N_G(K)$. Finally, K is normal in G if and only if $N_G(K) = G$, by Remark 3.1.3.

Proposition 3.1.5: Let H be a subgroup of G .

- a. Suppose $G = \langle S \rangle$ for some subset $S \subseteq G$. Then H is normal in G if and only if $\text{Inn}_x H = H$ for each $x \in S$.
- b. If $H = \langle T \rangle$ for some subset $T \subseteq H$, then H is normal in G if and only if $\forall t \in T, \forall x \in G, \text{Inn}_x t \in H$.

Proof:

a. ($\Rightarrow$): This follows from Remark 3.1.3.

($\Leftarrow$): The hypothesis says $S \subseteq N_G(H)$. Since $N_G(H)$ is a subgroup of G (see Definition 3.1.4), we get $G = \langle S \rangle \subseteq N_G(H)$. Thus $N_G(H) = G$, so H is normal in G .

b. ($\Rightarrow$): This implication follows from the definition of normal subgroup, since $T \subseteq H$.

($\Leftarrow$): Fix $x \in G$; we must argue that $\text{Inn}_x H \subseteq H$. Since Inn_x is a group homomorphism (see Proposition 3.1.1), it carries the subgroup generated by T onto the subgroup generated by $\text{Inn}_x(T)$:

$$\text{Inn}_x(\langle T \rangle) = \langle \text{Inn}_x(T) \rangle.$$

By assumption $\text{Inn}_x(T) \subseteq H$, and since H is a subgroup we get $\langle \text{Inn}_x(T) \rangle \subseteq H$. Thus $\text{Inn}_x H \subseteq H$. ■

Proposition 3.1.6:

Let $N = \ker \varphi$ where $\varphi : G \rightarrow H$ is a group homomorphism. Then N is a normal subgroup of G .

Proof: We have already observed that N is a subgroup. Now let $g \in G$ and $x \in N$ so that $\varphi(x) = 1$. Then

$$\varphi(\text{Inn}_g(x)) = \varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g^{-1}) = \varphi(g)\varphi(g)^{-1} = 1$$

so that $\text{Inn}_g N \subseteq N$ as required. ■

Corollary 3.1.7: The center $Z(G)$ is a normal subgroup of G .

Proof: $Z(G) = \ker(\text{Inn} : G \rightarrow \text{Aut}(G))$, since $\text{Inn}_g = \text{id}$ if and only if $gxg^{-1} = x$ for all $x \in G$. Now apply Proposition 3.1.6. ■

We will see below that in fact *every* normal subgroup is the kernel of some homomorphism.

Example 3.1.8:

Consider the group $\text{GL}_2(\mathbb{Q})$. For $x \in \mathbb{Q}$ write

$$\alpha(x) = \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix}.$$

Note for $x, y \in \mathbb{Q}$ that $\alpha(x+y) = \alpha(x)\alpha(y)$; thus $\alpha : \mathbb{Q} \rightarrow \text{GL}_2(\mathbb{Q})$ is an injective group homomorphism (with $\mathbb{Q}$ regarded as an additive group) whose image

$$U_{\mathbb{Q}} = \{\alpha(x) \mid x \in \mathbb{Q}\} = \left\{ \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix} \mid x \in \mathbb{Q} \right\}$$

is a subgroup of $\text{GL}_2(\mathbb{Q})$.

Observe that $U_{\mathbb{Z}} = \{\alpha(x) \mid x \in \mathbb{Z}\}$ is a subgroup of $U_{\mathbb{Q}}$.

For $t \in \mathbb{Q}^{\times} = \mathbb{Q} \setminus \{0\}$, write

$$h(t) = \begin{bmatrix} t & 0 \\ 0 & 1 \end{bmatrix}.$$

For $t, s \in \mathbb{Q}^{\times}$, we have $h(ts) = h(t)h(s)$ so that $h : \mathbb{Q}^{\times} \rightarrow \text{GL}_2(\mathbb{Q})$ is an injective group homomorphism whose image

$$T = \{h(t) \mid t \in \mathbb{Q}^\times\} = \left\{ \begin{bmatrix} t & 0 \\ 0 & 1 \end{bmatrix} \mid t \in \mathbb{Q}^\times \right\}$$

is a subgroup of $\mathrm{GL}_2(\mathbb{Q})$.

We observe for $t \in \mathbb{Q}^\times$ and $x \in \mathbb{Q}$ that

$$h(t)\alpha(x)h(t)^{-1} = \begin{bmatrix} t & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} t^{-1} & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & tx \\ 0 & 1 \end{bmatrix} = \alpha(tx).$$

This shows that $\forall s \in T, \mathrm{Inn}_s U_{\mathbb{Q}} \subseteq U_{\mathbb{Q}}$ so that T normalizes $U_{\mathbb{Q}}$.

Let $n \in \mathbb{Z}$ with $n > 1$ and consider the subgroup $C = C_n = \langle h(n) \rangle$ of T .

The generator $h(n)$ satisfies $\mathrm{Inn}_{h(n)} U_{\mathbb{Z}} \subseteq U_{\mathbb{Z}}$ since for $x \in \mathbb{Z}$

$$h(n)\alpha(x)h(n)^{-1} = \alpha(nx) \in U_{\mathbb{Z}}.$$

Note however that $\mathrm{Inn}_{h(n)} U_{\mathbb{Z}}$ is a proper subset of $U_{\mathbb{Z}}$; indeed, identifying $U_{\mathbb{Z}}$ with $\mathbb{Z}$, the image subgroup $\mathrm{Inn}_{h(n)} U_{\mathbb{Z}}$ identifies with $n\mathbb{Z}$ and of course $n\mathbb{Z}$ has index n in $\mathbb{Z}$.

The group $U_{\mathbb{Z}}$ is not normalized by $C = \langle h(n) \rangle$ since $\mathrm{Inn}_{h(n)^{-1}} U_{\mathbb{Z}} = \mathrm{Inn}_{h(n^{-1})} U_{\mathbb{Z}} \not\subseteq U_{\mathbb{Z}}$; indeed

$$\mathrm{Inn}_{h(n^{-1})} \alpha(1) = \alpha\left(\frac{1}{n}\right) = \begin{bmatrix} 1 & \frac{1}{n} \\ 0 & 1 \end{bmatrix} \notin U_{\mathbb{Z}}.$$

This example shows the following: there exists a group G , subgroups H, K of G , and a subset $S \subseteq H$ with $H = \langle S \rangle$, such that $\mathrm{Inn}_x K \subseteq K$ for all $x \in S$ even though H does not normalize K . (Take $G = \mathrm{GL}_2(\mathbb{Q})$, $K = U_{\mathbb{Z}}$, $H = C_n$, and $S = \{h(n)\}$.)

Of course, if we insist that $\mathrm{Inn}_x K = K$ for all $x \in S$ then H will normalize K ; see Proposition 3.1.5.

This example shows why the definition of the normalizer $N_G(K)$ in Definition 3.1.4 requires $\mathrm{Inn}_g K = K$ rather than merely $\mathrm{Inn}_g K \subseteq K$. With the weaker condition the resulting set need not be closed under inverses, and so need not be a subgroup.

Remark 3.1.9: The subgroup of $\mathrm{GL}_2(\mathbb{Q})$ generated by $h(n)$ and $\alpha(1)$ is isomorphic to $\mathbb{Z}\left[\frac{1}{n}\right] \rtimes \mathbb{Z}$; this is the *Baumslag–Solitar group* $\mathrm{BS}(1, n)$, which you might (?) meet again in a course on geometric group theory.

3.2. Quotient groups

Let N be a subgroup of G . We write G/N for the set of cosets gN , and $\pi = \pi_{G/N} : G \rightarrow G/N$ for the map $\pi(g) = gN$. Thus $\pi(g) = \pi(g') \Leftrightarrow g^{-1}g' \in N$; equivalently, $g' = gh$ for some $h \in N$.

Theorem 3.2.1: Let N be a normal subgroup of G . Then there is a unique group structure on G/N for which π is a group homomorphism. For this group structure:

a. the multiplication satisfies

$$\forall g, g' \in G, \pi(g)\pi(g') = \pi(gg')$$

b. the identity is given by $1_{G/N} = \pi(1_G)$,

c. inversion satisfies $\forall g \in G, \pi(g)^{-1} = \pi(g^{-1})$.

Moreover $\ker \pi = N$.

Proof: Uniqueness. Since π is surjective, any multiplication for which π is a homomorphism must satisfy 1.; this determines the multiplication, hence also the identity and inverses.

Existence. We must show that the rule in 1. gives a well-defined multiplication. Suppose $\pi(g) = \pi(g_1)$ and $\pi(g') = \pi(g'_1)$, so that $g = g_1 h$ and $g' = g'_1 h'$ with $h, h' \in N$. Then

$$gg' = g_1 h g'_1 h' = g_1 g'_1 \cdot (g'_1{}^{-1} h g'_1) h'.$$

Since N is normal, $g'_1{}^{-1} h g'_1 \in N$, so $(g'_1{}^{-1} h g'_1) h' \in N$ and thus $\pi(gg') = \pi(g_1 g'_1)$. This shows that

$$\mu(\pi(g), \pi(g')) = \pi(gg')$$

is a well-defined map $\mu : G/N \times G/N \rightarrow G/N$.

We check the group axioms. Let $z, z', z'' \in G/N$; since π is surjective we may write $z = \pi(g)$, $z' = \pi(g')$, $z'' = \pi(g'')$. Then

$$(zz')z'' = \pi(gg')\pi(g'') = \pi((gg')g'') = \pi(g(g'g'')) = \pi(g)\pi(g'g'') = z(z'z'').$$

Next, $\pi(1)\pi(g) = \pi(g) = \pi(g)\pi(1)$, so $\pi(1)$ is an identity element, and $\pi(g)\pi(g^{-1}) = \pi(1) = \pi(g^{-1})\pi(g)$ shows that $\pi(g)^{-1} = \pi(g^{-1})$. Thus G/N is a group, and 1. says precisely that π is a homomorphism.

Finally, $\pi(g) = \pi(1)$ if and only if $g \in N$, so $\ker \pi = N$. ■

Corollary 3.2.2: A subgroup of G is normal if and only if it is the kernel of some group homomorphism with domain G .

Proof: A kernel is normal by Proposition 3.1.6. Conversely, if N is normal then $N = \ker \pi_{G/N}$ by Theorem 3.2.1. ■

Example 3.2.3:

If G is an abelian group, then Inn_x is the trivial homomorphism for each $x \in G$, and in particular every subgroup of G is normal.

Let's consider an additive abelian group A and B any subgroup. Write $\pi : A \rightarrow A/B$ for the quotient mapping.

For $a \in A$, we often view $\pi(a)$ as the coset $a + B = \{a + x \mid x \in B\}$.

We see for $a, a' \in A$ that $\pi(a) = \pi(a') \Leftrightarrow a - a' \in B$.

Example 3.2.4: A non-example. Let $G = S_3$ and $H = \langle (1\ 2) \rangle$, which is not normal. Then $\pi(1) = \pi((1\ 2))$, but $\pi((1\ 3)) \neq \pi((1\ 2)(1\ 3))$: indeed, $(1\ 3)^{-1}(1\ 2)(1\ 3) = (2\ 3) \notin H$. So the product of cosets is not well-defined: the argument in the proof above breaks down exactly where normality was used.

3.3. First isomorphism theorem

Theorem 3.3.1:

Let $\varphi : G \rightarrow H$ be a group homomorphism, and let $K = \ker \varphi$. Then there is a unique group homomorphism $\bar{\varphi} : G/K \rightarrow H$ such that $\varphi = \bar{\varphi} \circ \pi$, where $\pi : G \rightarrow G/K$ is the quotient homomorphism. Moreover $\bar{\varphi}$ is injective with image $\text{im}(\varphi)$, so that $\bar{\varphi}$ induces an isomorphism $G/K \simeq \text{im}(\varphi)$. In particular, if φ is surjective then $\bar{\varphi} : G/K \rightarrow H$ is an isomorphism.

Proof: Note that K is normal by Proposition 3.1.6, so G/K is a group.

Uniqueness. For any $z \in G/K$ we may write $z = \pi(g)$ for $g \in G$, and then the requirement $\varphi = \bar{\varphi} \circ \pi$ forces

$$(*) \quad \bar{\varphi}(z) = \bar{\varphi}(\pi(g)) = \varphi(g).$$

Well-defined. We must check that $(*)$ does not depend on the choice of g . If $\pi(g) = \pi(g')$ then $g' = gk$ with $k \in K$, so $\varphi(g') = \varphi(g)\varphi(k) = \varphi(g)$.

Homomorphism. For $z, z' \in G/K$ with $z = \pi(g)$ and $z' = \pi(g')$ we have

$$\bar{\varphi}(zz') = \bar{\varphi}(\pi(gg')) = \varphi(gg') = \varphi(g)\varphi(g') = \bar{\varphi}(z)\bar{\varphi}(z').$$

Image. Since $\varphi = \bar{\varphi} \circ \pi$ and π is surjective, $\text{im}(\bar{\varphi}) = \text{im}(\varphi)$.

Injective. It suffices to show that $\ker \bar{\varphi}$ is trivial; see Proposition 1.2.5. Let $z \in \ker \bar{\varphi}$ and write $z = \pi(g)$. Then $1_H = \bar{\varphi}(\pi(g)) = \varphi(g)$, so $g \in K$, and hence $z = \pi(g) = \pi(1) = 1_{G/K}$. ■

3.4. Groups acting on Groups

Let G and H be groups and suppose that G acts on the set H .

Definition 3.4.1:

We say that G acts by automorphisms on H if for each $g \in G$, the mapping

$$h \mapsto g \cdot h : H \rightarrow H$$

is an automorphism of the group H .

Remark 3.4.2:

To give an action of G on H by automorphisms is the same as to give a group homomorphism $G \rightarrow \text{Aut}(H)$.

Proposition 3.4.3:

If G acts on H by automorphisms, the set of fixed points

$$H^G = \{x \in H \mid \forall g \in G, g \cdot x = x\}$$

is a subgroup of H .

Proof: Each automorphism ψ of H satisfies $\psi(1) = 1$, so $1 \in H^G$.

Let $x, y \in H^G$ and $g \in G$. Since g acts by an automorphism of H ,

$$g \cdot (x^{-1}y) = (g \cdot x)^{-1}(g \cdot y) = x^{-1}y,$$

so $x^{-1}y \in H^G$. Thus H^G is a subgroup. ■

Example 3.4.4:

G acts on itself by inner automorphisms. This action is determined by the group homomorphism $\text{Inn} : G \rightarrow \text{Aut}(G)$.

In this case, the subgroup G^G of fixed points is precisely the center $Z = Z(G)$:

$$Z = \{x \in G \mid \text{Inn}_g x = x \quad \forall g \in G\} = \{x \in G \mid gx = xg \quad \forall g \in G\}$$

For $x \in G$, the stabilizer $\text{Stab}_G(x)$ is known as the centralizer:

$$\text{Stab}_G(x) = C_G(x) = \{g \in G \mid \text{Inn}_g x = x\} = \{g \in G \mid gx = xg\}.$$

And the orbit of x is known as the conjugacy class of x :

$$\mathcal{O}_x = \text{Cl}(x) = \{\text{Inn}_g x \mid g \in G\} = \{gxg^{-1} \mid g \in G\}.$$

Proposition 2.5.3 gives a bijection

$$\text{Cl}(x) \simeq G/C_G(x).$$

4. Week 4 [2026-09-28]

4.1. p -groups

Definition 4.1.1:

For a prime number p , a finite p -group is a finite group G whose order is a power of p .

Let G be a finite p -group and suppose that G acts on the finite set E , and write E^G for the set of elements of E fixed by the action of G ; thus $E^G = \{x \in E \mid \forall g \in G, g \cdot x = x\}$.

Proposition 4.1.2:

With notation as above, we have $|E| \equiv |E^G| \pmod{p}$.

Proof: The set E is the disjoint union of the G -orbits. For $x \in E$, the orbit of x has size $[G : \text{Stab}_G(x)]$, which divides $|G| = p^k$ and so is a power of p ; it equals 1 exactly when $x \in E^G$. Thus the complement $E \setminus E^G$ is a disjoint union of orbits, each of size divisible by p , so $|E| - |E^G|$ is divisible by p . ■

Proposition 4.1.3:

Let G be a finite p -group, and suppose that G acts by automorphisms on a non-trivial finite p -group H (for the same prime p). Then the fixed points H^G form a non-trivial subgroup of H .

Proof: H^G is a subgroup by Proposition 3.4.3. By Proposition 4.1.2 (applied with $E = H$) we have $|H^G| \equiv |H| \equiv 0 \pmod{p}$, since H is a non-trivial p -group. Since $1 \in H^G$, we have $|H^G| \geq 1$; being divisible by p , we get $|H^G| \geq p$. So H^G is non-trivial. ■

Theorem 4.1.4: The center of a non-trivial finite p -group is non-trivial.

Proof: If G is a non-trivial p -group, consider the action of G on itself by conjugation; this is an action by automorphisms. The subgroup of fixed points is precisely the center of G , and Proposition 4.1.3 (with $H = G$) implies that this subgroup is non-trivial. ■

Corollary 4.1.5:

Let G be a finite p -group with $|G| = p^n$. There is a series of subgroups

$$\{1\} = G_n \subset G_{n-1} \subset \dots \subset G_0 = G$$

such that $|G_i| = p^{n-i}$, such that G_i is normal in G for each $0 \leq i \leq n$, and such that G_i/G_{i+1} is cyclic of order p for $0 \leq i < n$.

Proof: We proceed by induction on n . If $n = 0$ then G is trivial and there is nothing to prove.

Now suppose $n \geq 1$ and that the result holds for all p -groups of order $< |G|$.

By Theorem 4.1.4 the center Z of G is non-trivial. Choose $x \in Z$ with $x \neq 1$; its order is p^k for some $k \geq 1$. Then $z = x^{p^{k-1}}$ has order p , and $C = \langle z \rangle$ is a subgroup of Z of order p . Every subgroup of Z is normal in G , since Inn_G is trivial on Z . Let $\pi : G \rightarrow G/C$ be the quotient homomorphism. Then $|G/C| = p^{n-1}$, so by induction there is a series

$$\{1\} = H_{n-1} \subset H_{n-2} \subset \dots \subset H_0 = G/C$$

of normal subgroups of G/C with $|H_i| = p^{n-1-i}$ and H_i/H_{i+1} cyclic of order p .

Define $G_i = \pi^{-1}(H_i)$ for $0 \leq i \leq n-1$, and $G_n = \{1\}$. Then:

- $|G_i| = |H_i| \cdot |C| = p^{n-i}$ for $0 \leq i \leq n-1$, and $G_0 = G$.
- G_i is normal in G : for $g \in G$ and $x \in G_i$ we have $\pi(gxg^{-1}) = \pi(g)\pi(x)\pi(g)^{-1} \in H_i$.
- For $0 \leq i \leq n-2$, the composite $G_i \rightarrow H_i \rightarrow H_i/H_{i+1}$ (restriction of π , followed by the quotient map) is surjective with kernel G_{i+1} . By the first isomorphism theorem, $G_i/G_{i+1} \simeq H_i/H_{i+1}$, which is cyclic of order p .
- $G_{n-1} = \pi^{-1}(\{1\}) = C$, so $G_{n-1}/G_n \simeq C$ is cyclic of order p .

This gives the desired series. ■

4.2. Sylow subgroups

Let G be a finite group of order $n = p^m q$ with p a prime and with $\gcd(p, q) = 1$.

Theorem 4.2.1 (Sylow's Theorem):

There exists a subgroup of G having order p^m ; such a subgroup is known as a *Sylow subgroup*, or a *Sylow p -subgroup*. Moreover:

- Any two Sylow p -subgroups are conjugate by an element of G .
- Any p -subgroup of G is contained in a Sylow p -subgroup.
- If r denotes the number of p -Sylow subgroups, then $r \equiv 1 \pmod{p}$ and $r \mid q$.

For the proof, we consider the set E of all subsets of G having order p^m . The action of G on itself by translation induces an action of G on E : for $X \in E$, evidently $g \cdot X \in E$ where $g \cdot X = \{g \cdot x \mid x \in X\}$.

One knows that $|E| = \binom{|G|}{p^m} = \binom{p^m q}{p^m}$.

Proposition 4.2.2: $\binom{p^m q}{p^m} \equiv q \pmod{p}$.

Proof: Let X and Y be indeterminants; we work in the polynomial ring $(\mathbb{Z}/p\mathbb{Z})[X, Y]$. Write $n = p^m q$ and consider

$$(X + Y)^n = ((X + Y)^{p^m})^q = (X^{p^m} + Y^{p^m})^q = \sum_{i=0}^q \binom{q}{i} (X^{p^m})^i (Y^{p^m})^{q-i}.$$

On the other hand, we have

$$(X + Y)^n = \sum_{i=0}^n \binom{n}{i} X^i Y^{n-i}.$$

and the required result follows by comparing the coefficient of $X^{p^m} Y^{(q-1)p^m}$ in the two expressions. ■

Definition 4.2.3: Let G be a group and let $H \subseteq G$ be a subgroup. The **normalizer of H in G** is the subgroup

$$N_G(H) = \{g \in G \mid \text{Inn}_g H = H\};$$

it is the stabilizer of H in G for the conjugation action of G on the set of subgroups of G .

Notice that $G/N_G(H)$ is in bijection with the set of all conjugates $\{gHg^{-1} \mid g \in G\}$.

For the proof of the Theorem, we are going to use the following:

Proposition 4.2.4: Let P be a Sylow p -subgroup of G and let Q be any p -subgroup of G . Then

$$N_Q(P) = Q \cap P.$$

Proof: By definition, $N_Q(P) = Q \cap N_G(P)$, so we must show that $Q \cap N_G(P) = Q \cap P$.

Let $H = Q \cap N_G(P)$. Since $P \subseteq N_G(P)$, it is clear that $Q \cap P \subseteq H = Q \cap N_G(P)$. It remains to establish the reverse inclusion. Since $H \subseteq Q$ by definition, it only remains to prove that $H \subseteq P$.

For this, we first claim that PH is a p -subgroup of G . Assume for the moment that this claim has been established. Since PH contains P and since P is a p -subgroup of maximal possible order, we conclude that $P = PH$ and hence that $H \subseteq P$ as required.

Since $H \subseteq N_G(P)$, the product $PH = \{xh \mid x \in P \text{ and } h \in H\}$ is a subgroup of G . Moreover, we've seen for homework that $|PH| = \frac{|P||H|}{|P \cap H|}$. Since $|P|$ and $|H|$ are powers of p , PH is a p -subgroup. ■

Finally, we now give:

Proof of Sylow's Theorem: Proposition 4.2.2 shows that $|E| \not\equiv 0 \pmod{p}$. Thus there must be some $X \in E$ for which the orbit $G \cdot X$ satisfies $|G \cdot X| \not\equiv 0 \pmod{p}$. If H is the stabilizer in G of X , there is of course a bijection between $G \cdot X$ and G/H . In particular,

$$|G/H| \not\equiv 0 \pmod{p}.$$

Since $|G| = |H| \cdot |G/H|$, conclude that p^m divides the order of H .

On the other hand, fix $x \in X$. We claim that $H \subseteq X \cdot x^{-1}$. Indeed, for $h \in H$, since h stabilizes X we have

$$hx = x' \text{ for some } x' \in X.$$

Then $h = x'x^{-1} \in X \cdot x^{-1}$ as required.

Concluding, we find that $|H| \leq |X \cdot x^{-1}| = |X| = p^m$ and thus $|H| = p^m$. In particular, H is a Sylow subgroup.

Now let H' be any p -subgroup of G and consider the action of H' on the quotient G/H determined by left-multiplication. Since $|G/H| = q$ is not divisible by p , Proposition 4.1.2 shows that $(G/H)^{H'} \neq \emptyset$. Suppose that the coset $gH \in G/H$ is fixed by H' . We claim that

$$H' \subset gHg^{-1}.$$

Indeed, since gH is fixed by H' , we have

$$x \in H' \Rightarrow xgH = gH \Rightarrow g^{-1}xgH = H \Rightarrow g^{-1}xg \in H \Rightarrow x \in gHg^{-1}.$$

This confirms that $H' \subseteq gHg^{-1}$. Thus any p -subgroup of G is contained in a Sylow subgroup. This proves (b).

Applying the argument of the preceding paragraph to the case where H' is a Sylow subgroup we see that $H' = gHg^{-1}$; this shows that any two Sylow subgroups are conjugate, proving (a).

To prove (c), let P be a Sylow p -subgroup. Note that P acts by conjugation on the set of all Sylow p -subgroups of G . We choose Sylow p -subgroups $Q_1, Q_2, \dots, Q_s$ which form a system of representatives of the P -orbits for this action. We may and will take $Q_1 = P$.

For $1 \leq i < s$, we write $\mathcal{O}_i = P \cdot Q_i = \{xQ_ix^{-1} \mid x \in P\}$ for the P -orbit of Q_i . Recall that $\mathcal{O}_i$ is in bijection with the quotient $P/N_P(Q_i)$ where $N_P(Q_i)$ is the *normalizer* of Q_i in P .

For $1 \leq i \leq s$ Proposition 4.2.4 shows that $N_P(Q_i) = Q_i \cap P$.

In particular, it follows that $N_P(Q_1) = P \cap P = P$ so that $|\mathcal{O}_1| = 1$. For all $2 \leq i \leq s$ we have $P \neq Q_i$ so that $N_P(Q_i) = Q_i \cap P \subsetneq P$. Thus $|\mathcal{O}_i| = [P : Q_i \cap P] > 1$ so that

$$|\mathcal{O}_i| \equiv 0 \pmod{p}.$$

Finally, the number r of Sylow p -subgroups satisfies

$$r = \sum_{i=1}^s |\mathcal{O}_i| = 1 + \sum_{i=2}^s |\mathcal{O}_i| \equiv 1 \pmod{p}$$

which proves the first assertion of (c). The second assertion of (c) follows since

$$r = [G : N_G(P)]$$

and since $P \subseteq N_G(P)$. ■

For a finite group G and a prime number p , write $n_p = n_p(G)$ for the number of p -Sylow subgroups of G (this is the number r from Theorem 4.2.1).

Corollary 4.2.5:

Let $P \in \text{Syl}_p(G)$. Then P is normal if and only if $n_p = 1$.

Proof: Indeed, P is normal if and only if $\text{Inn}_g P = P$. Since all p -Sylow subgroups are conjugate, the result is immediate. ■

4.3. Applications of Sylow's Theorem

Definition 4.3.1:

Let G be a group. A subgroup H of G is **characteristic** if for every automorphism $\varphi : G \rightarrow G$, we have $\varphi(H) = H$.

A characteristic subgroup H is always normal, since H is invariant under all the inner automorphisms Inn_g for $g \in G$.

Example 4.3.2:

For a prime number p , let $G = \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$; we may identify the group G with the quotient of $\mathbb{Z} \times \mathbb{Z}$ by the subgroup $N = \langle (p, 0), (0, p) \rangle$. Write $e = (1, 0) + N$ and $f = (0, 1) + N$ so that $e, f \in G$ are elements of order p and

$$G = \langle e, f \rangle = \langle e \rangle + \langle f \rangle \text{ and } \langle e \rangle \cap \langle f \rangle = \{0\}.$$

Since G is commutative, $H = \langle e \rangle$ is a normal subgroup of G . But H is not characteristic in G . Indeed, the automorphism $(a, b) \mapsto (b, a)$ of $\mathbb{Z} \times \mathbb{Z}$ induces an automorphism φ of $G = (\mathbb{Z} \times \mathbb{Z})/N$ for which

$$\varphi(e) = f \text{ and } \varphi(f) = e.$$

Thus $\varphi(H) \neq H$.

Definition 4.3.3:

A group G is **simple** if for any normal subgroup $N \subseteq G$, either $N = \{1\}$ or $N = G$.

Any group of prime order is simple. Below we are going to find the first example of a non-abelian simple group.

Throughout the remainder of this section, G denotes a finite group.

For a prime number p we write $n_p = n_p(G)$ for the number of p -Sylow subgroups in G , and $\text{Syl}_p = \text{Syl}_p(G)$ for the set of Sylow p -subgroups. Recall that

$$n_p \equiv 1 \pmod{p} \text{ and } p \mid |G/P| \text{ for } P \in \text{Syl}_p(G)$$

.

Proposition 4.3.4:

- If $n_p = 1$ for some prime p , then $P \in \text{Syl}_p(G)$ is characteristic – and in particular normal – in G .
- Suppose that $H \subseteq G$ is a normal subgroup, and suppose that $P \in \text{Syl}_p(H)$ is a normal p -Sylow subgroup of H for some prime p . Then P is a normal subgroup of G .

Proof:

- For any automorphism φ of G , $\varphi(P)$ is again a p -Sylow subgroup of G . Since $n_p = 1$, we then $\varphi(P) = P$ so that P is indeed characteristic.
- For $g \in G$, Inn_g determines an automorphism of H . Thus by (a), $\text{Inn}_g P = P$ which shows that P is normal in G .

■

Proposition 4.3.5:

Suppose that $|G| = pq$ for distinct prime numbers $p < q$.

- $n_q = 1$ so that G has a normal Sylow q -subgroup.
- If p does not divide $q - 1$ then $n_p \equiv 1 \pmod{p}$ so that G has a normal Sylow p -subgroup. In this case, G is a cyclic group.

Proof:

- By Theorem 4.2.1 we have $n_q \equiv 1 \pmod{q}$ and $n_q \mid p$. Since $q > p$ it follows that $n_q = 1$.
- Again we have $n_p \equiv 1 \pmod{p}$ and $n_p \mid q$. Since q is prime, the only possibilities are $n_p = 1$ or $n_p = q$. If $n_p = q$ then $q \equiv 1 \pmod{p}$ so that p divides $q - 1$.

If $n_p = 1$ then $G = PQ$ where P is a Sylow p -subgroup and Q is a Sylow q -subgroup. You will prove for homework that since P and Q are both normal in G , G is isomorphic to the direct product $P \times Q$. Thus

$$G \simeq \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z} \simeq \mathbb{Z}/pq\mathbb{Z}$$

is indeed cyclic. ■

Proposition 4.3.6:

Suppose that the size of a p -Sylow subgroup of G is p . Then G contains exactly $(p-1) \cdot n_p$ elements of order p .

Proof: Indeed, suppose that $P, Q \in \text{Syl}_p(G)$ with $P \neq Q$. Since $|P| = |Q| = p$ is prime we know that $P \cap Q = \{1\}$. Thus

$$\left| \bigcup_{P \in \text{Syl}_p} P \setminus \{1\} \right| = \sum_{P \in \text{Syl}_p} (p-1) = n_p(p-1).$$

Proposition 4.3.7:

Suppose that $|G| = 12$. Then either $n_3 = 1$ or G is isomorphic to the alternating group A_4 , and in that case, $n_2 = 1$.

Proof:

We know that $n_3 \equiv 1 \pmod{3}$ and $n_3 \mid 4$. Thus n_3 is either 1 or 4.

To complete the proof we must show that if $n_3 = 4$ then G is isomorphic to A_4 .

The group G acts by conjugation on the set $\Omega = \text{Syl}_3$ of 3-Sylow subgroups, and $|\Omega| = 4$. This action determines a group homomorphism

$$\varphi : G \rightarrow S(\Omega) = S_4.$$

The kernel K of φ consists of all $g \in G$ which normalize each Sylow 3-subgroup. In particular for $P \in \Omega$, $K \subseteq N_G(P) = P$. Since K is normal, since P is not normal, and since $|P| = 3$, we conclude that $K = \{1\}$.

Thus φ is an isomorphism from G to its image in S_4 .

Since $n_3 = 4$, Proposition 4.3.6 shows that G contains exactly 8 elements of order 3. On the other hand, there are exactly 8 three-cycles in S_4 and all are contained in A_4 .

Thus the intersection of the image of φ with A_4 is a group containing at least 8 elements. Since both groups have order 12, they must coincide.

Finally, notice that $K = \langle (12)(34), (13)(24) \rangle$ is a normal 2-sylow subgroup of A_4 ; thus $n_2(A_4) = 1$. ■

Proposition 4.3.8:

Suppose that $|G| = 30$. Then $n_5 = 1$ so that G has a normal Sylow 5-subgroup.

Proof: Let $P \in \text{Syl}_5$ and $Q \in \text{Syl}_3$.

First suppose that neither P nor Q is normal in G . In that case, $n_5 = 6$ and $n_3 = 10$. By counting elements of order 3 and of order 5, it now follows from Proposition 4.3.6 that

$$|G| \geq 6 \cdot 4 + 10 \cdot 2 = 44 > 30.$$

This contradiction proves that at least one of P or Q is normal in G .

Now if either P or Q is normal in G , then PQ is a subgroup of order 15. It follows from Proposition 4.3.5 that any group of order 15 is cyclic. Using Proposition 4.3.4 we then conclude that both P and Q are normal in G , and the result follows. ■

Proposition 4.3.9:

Suppose that $|G| = 60$. If $n_5 > 1$ then G is a simple group.

Proof: For any group of order 60, n_5 is either 1 or 6; thus we suppose $n_5 = 6$.

Let $P \in \text{Syl}_5(G)$. If $N_G(P)$ is the normalizer of P , then $|G/N_G(P)| = 6$ so that $|N_G(P)| = 10$.

We proceed by contradiction; thus, we suppose that $\{1\} \neq H \subsetneq G$ is a normal subgroup of G .

First suppose that $5 \mid |H|$. Then H contains a Sylow 5-subgroup of G ; since H is normal, H contains all six Sylow 5-subgroups of G . Counting elements of order 5 in H , it follows from Proposition 4.3.6 that

$$|H| \geq 6 \cdot 4 = 24.$$

Since the only divisor d of 60 with $d \geq 24$ is $d = 30$, we conclude that $|H| = 30$. Now Proposition 4.3.8 shows that H has a normal 5-Sylow subgroup $Q \in \text{Syl}_5(H)$, and Proposition 4.3.4 shows that Q is normal in G . But this contradicts the assumption $n_5 > 1$.

This shows that

$$(\clubsuit) \quad G \text{ has no normal subgroup } H \text{ for which } 5 \mid |H|.$$

Thus we may suppose that $|H|$ is a divisor of $60/5 = 12$.

If $|H| = 12$, it follows from Proposition 4.3.7 that G has a normal Sylow p -subgroup for either $p = 2$ or $p = 3$. In view of Proposition 4.3.4, it follows that G has a normal subgroup of order 4 or 3.

If $|G| = 6$, then G has a normal Sylow 3-subgroup by Proposition 4.3.5.

Thus we may suppose that $|H|$ is one of 2, 3, 4.

Write $\overline{G} = G/H$ for the quotient group, so that $\overline{G} = 30, 20$ or 15 .

We claim in each case that $\overline{G}$ has a normal subgroup $\overline{Q}$ of order 5, i.e. that $n_5(\overline{G}) = 1$.

If $|\overline{G}| = 30$ this claim follows from Proposition 4.3.8. If $|\overline{G}| = 20$ note that $n_5 \mid 4$ and $n_5 \equiv 1 \pmod{5}$ shows that $n_5 = 1$. Finally, if $|\overline{G}| = 15$, then n_5 divides 3 and $n_5 \equiv 1 \pmod{5}$ again shows that $n_5 = 1$.

If $\pi : G \rightarrow \overline{G} = G/H$ is the quotient mapping, let $H_1 = \pi^{-1}(\overline{Q})$ be the inverse image of the normal subgroup $\overline{Q}$ of order 5. You will prove for homework that H_1 is a normal subgroup of G containing H ; since $H_1/H \simeq \overline{Q}$ it follows that $5 \mid |H_1|$. This contradicts $(\clubsuit)$ and completes the proof of the Proposition. ■

Corollary 4.3.10:

The alternating group A_5 is a simple group of order 60.

Proof: Indeed, the subgroups $\langle(1, 2, 3, 4, 5)\rangle$ and $\langle(1, 3, 2, 4, 5)\rangle$ are two distinct 5-Sylow subgroups of A_5 so that $n_5(A_5) > 1$. ■